Course Code: DF101 – Basic Computer Skills for Digital Forensics.

Lab Title: Lab 1B: PC System Fundamentals, Disk Sectors and Partition Analysis

Full name: Esther Kubeh Awuku

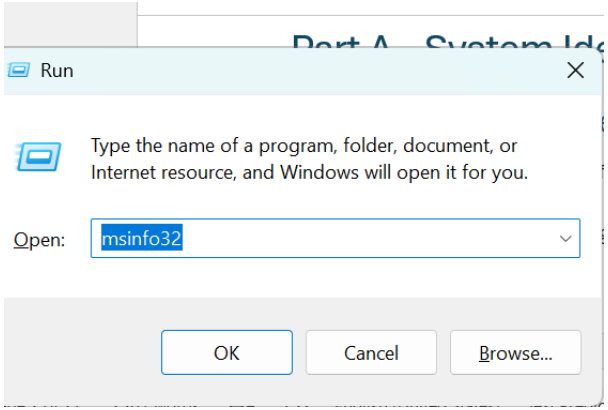
Platform Used: Kali Linux

Path: C:\Digital_Forensics_Lab

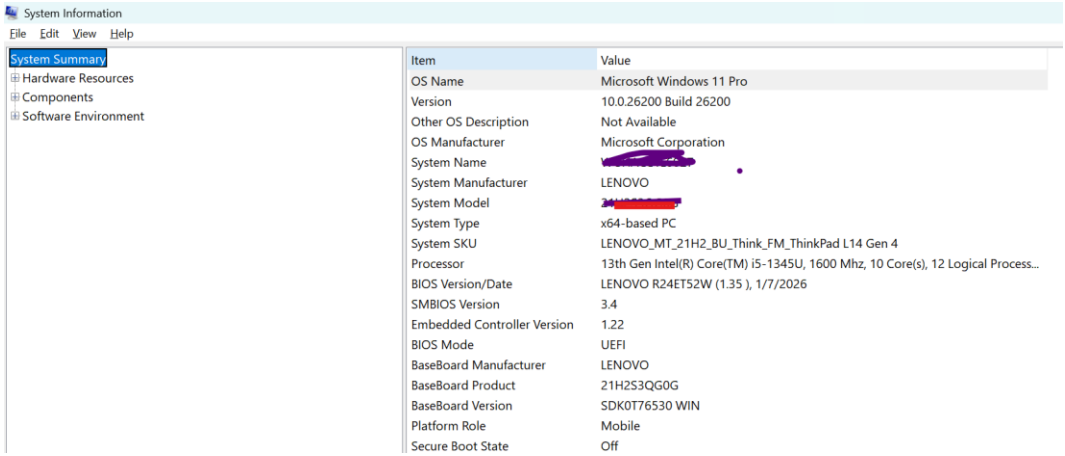
Part A - System Identification and Evidence Layer Mapping

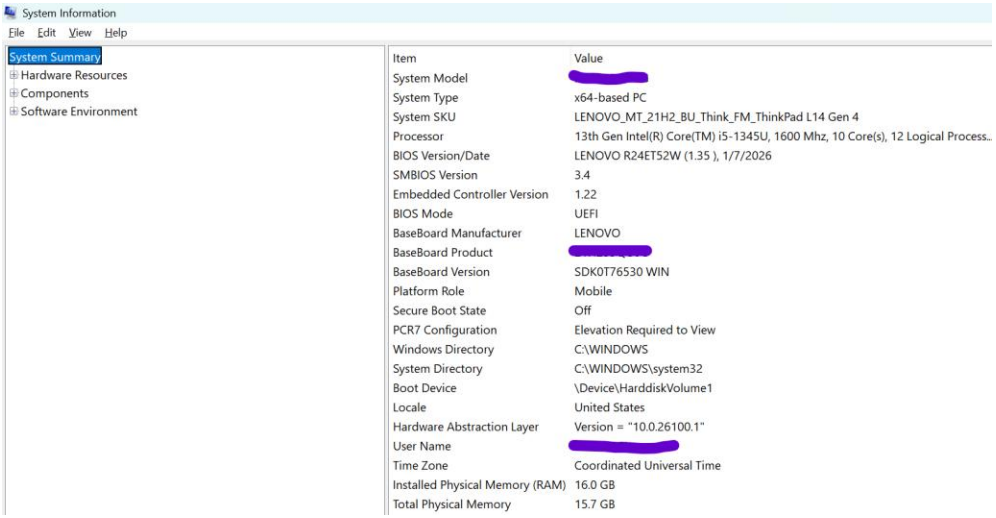
A1. Capture basic system specification

- On Windows, open System Information by pressing Windows + R, typing msinfo32, and pressing Enter.



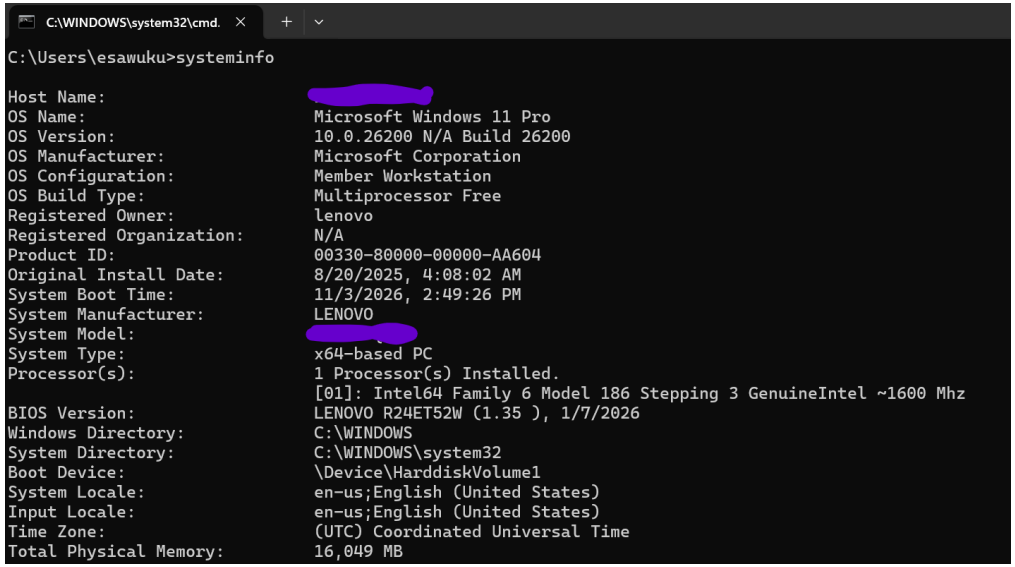
- Capture a screenshot showing OS Name, Version, System Type, Processor, BIOS Mode, Installed Physical Memory, and Secure Boot State if available.





- Also run the command below in Command Prompt and capture the first page of output.

```
systeminfo
```



A2. Capture storage information

- Open Command Prompt or PowerShell.
- Run one of the command sets below. Capture screenshots of storage model, size, partitions, and file systems.
- If a command is not supported on your system, use the alternative command.

```
wmic diskdrive get model,size,serialnumber
wmic logicaldisk get caption,filesystem,size,freespace,volumename

PowerShell alternative:
Get-Disk
Get-Partition
Get-Volume
```

Screenshot “Get-Disk”



Screenshot Get-Partition

```
PS C:\Users\esawuku> Get-Partition


   DiskPath: \\?\scsi#disk&ven_nvme&prod_skhynix_hfs256ge#5&afc12e1&0&000000#{53f56307-b6bf-11d0-94f2-00a0c91efb8b}

PartitionNumber  DriveLetter Offset                                        Size Type
---------------  ----------- ------                                        ---- ----
1                            1048576                                    260 MB System
2                            273678336                                   16 MB Reserved
3                C           290455552                                 236.25 GB Basic
4                            253963010048                                1.95 GB Recovery
```

Screenshot Get-Volume

```
PS C:\Users\esawuku> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
            WinRE_DRV    NTFS           Fixed     Healthy      OK                    947.17 MB   1.95 GB
C           Windows      NTFS           Fixed     Healthy      OK                     28.02 GB 236.25 GB
```

A3. Map evidence sources to system layers

- Create a table in your report showing at least five system layers/sources and the type of evidence that can be generated from each one.
- Use this guide: User activities = Applications + Operating System + Disk/Memory.

Layer / Source	Example Evidence	Where It May Be Found	Why It Matters
User layer	Opened files, searched terms, typed usernames,	Recent files, browser history, shell history, can be found in windows recent item files and the user profiles	Links user action to system activity

	Confidential files recently accessed		
Application layer	Office document metadata, browser cache, chat logs, browser history	Application folders and databases, browser history data	Shows what software created or used evidence
OS layer	Event logs, installed programs, user profiles, user login records	Windows logs, registry, Linux /var/log, windows security logs	Shows system events and account activity
Disk layer	Deleted files, file fragments, partitions, file creation	File system and unallocated space, NTFS file-system metadata	Supports recovery and timeline analysis
Memory/process layer	Running malware, PIDs, network connections	RAM capture, process list, PowerShell process list	Shows live system behavior

- Explain why evidence may exist in more than one layer.

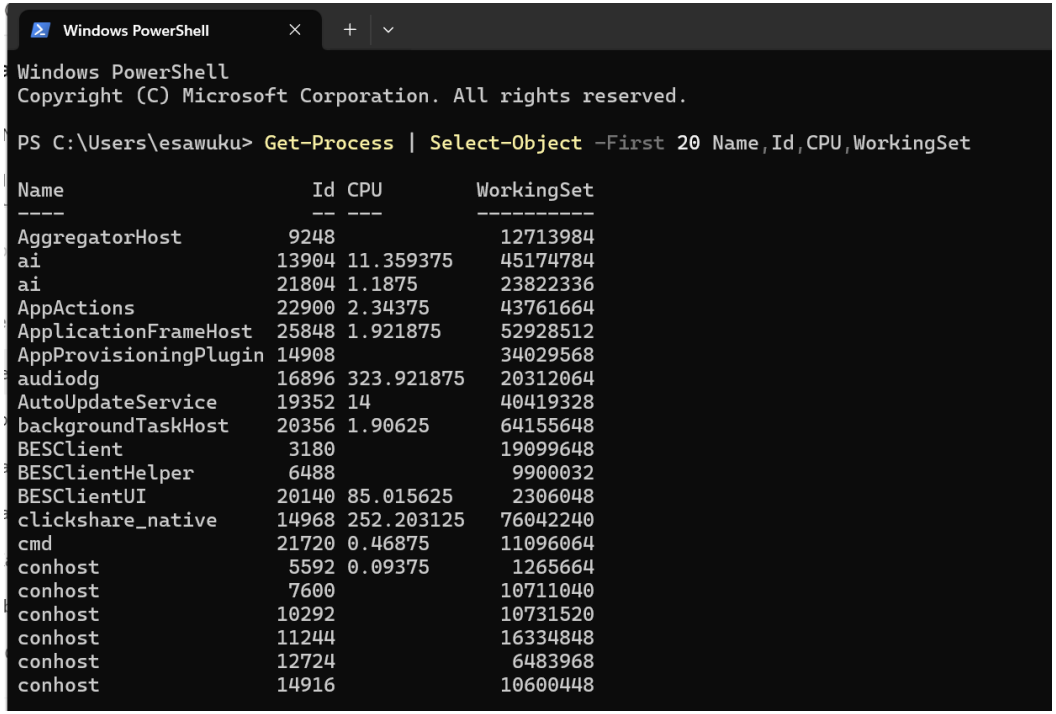
Evidence may exist in one or more layer because a user's activity can generate different kinds of records across multiple layers on the device. This process can help forensic investigators to analyze the data across the multiple layers to come up with a solution. An example could be a user visiting and browsing a website on Microsoft edge or chrome, the browser been used will keep some browsing history of the website, windows on the OS layer may also keep some user activities and the website information could also be stored on the browser's cache on the disk at the disk layer. So even tho we have one evidence, it exists in more than one layer.

❖ Part B - Program, Process and Thread Investigation

B1. Observe running processes

- Open Task Manager on Windows, or use Command Prompt/PowerShell.

- Capture a screenshot showing running processes.



- Select three processes and record their process names, PID values, and likely purpose.

Process Name	**PID**	**Likely Purpose**
audiodg	16896	This handles the audio processing for both windows and applications by managing the sound playback and audio effects
ApplicationFrameHost	25848	This process helps by supporting the display of certain Windows applications in their application windows

conhost	5592	When it comes to command-line applications like command prompt, this process supports that.

B2. Explain the difference

- In your report, explain the difference between a program, a process, and a thread in your own words.

A program is an application or software that is installed on a device containing some instructions for performing a particular task. For example, Microsoft excel

A process can be defined as a program that has been opened to be worked on and is currently running in the background. For example, when you open Microsoft word to work on something, the application starts running at the background to ensure smooth access. It normally runs in the Task Manager.

A thread is a small executable unit found in the process that performs a particular task. For example, when Microsoft excel is opened, different executable units called threads handle different parts of the process, like background, interactions etc.

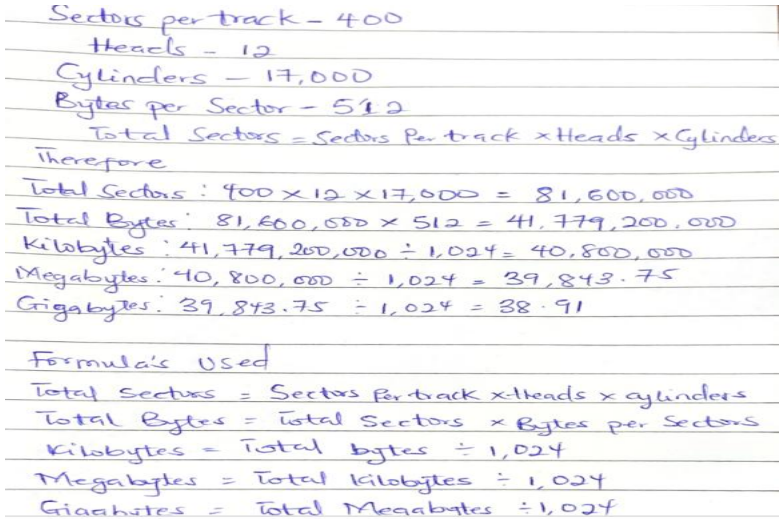
- Explain why process information may be useful during malware analysis or live forensic triage

Process information is quite useful when conducting malware analysis and live forensic triage, this is because it helps the forensic investigators to be able to identify the programs that are currently running on a computer. It helps in revealing unfamiliar or suspicious processes, any unusual resource usage and possible network activity. With process names, PIDs Investigators can identify and examine specific running programs. With this process information, some potential threats can be determined and investigated into.

Part C - Disk Sector and Disk Size Calculation

C1. Reproduce the slide calculation

Item	Formula / Value	Answer
Total sectors	400 x 12 x 17000	81,600,000 sectors
Total bytes	81,600,000 x 512	41,779,200,000 bytes
Kilobytes	41,779,200,000 / 1024	40,800,000 KB
Megabytes	40,800,000 / 1024	39,843.75 MB
Gigabytes	39,843.75 / 1024	38.91 GB

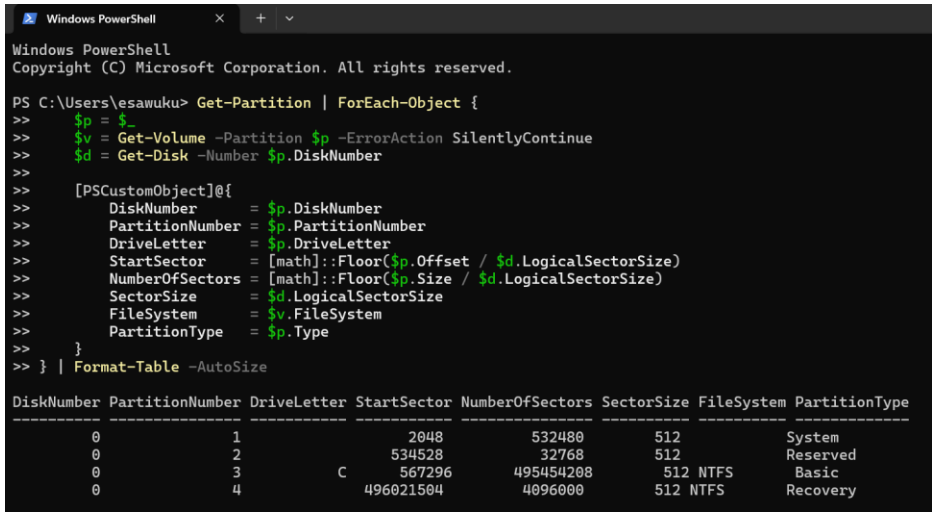


C2. Calculate your own disk or partition size

- Use your partition table or disk details to identify at least two partitions.
    - Partition 3 – Windows (‘C): Per my analysis, Partition 3 that is recommended, because it contains the windows operating system, application data of the system and other digital evidence that are very useful when it comes to forensic analysis.

    - Partition 4 – Recovery: I choose this partition because it contains all windows recovery tools. This recovery details can help forensic investigators when it comes to deleted or old information that could help in their analysis.

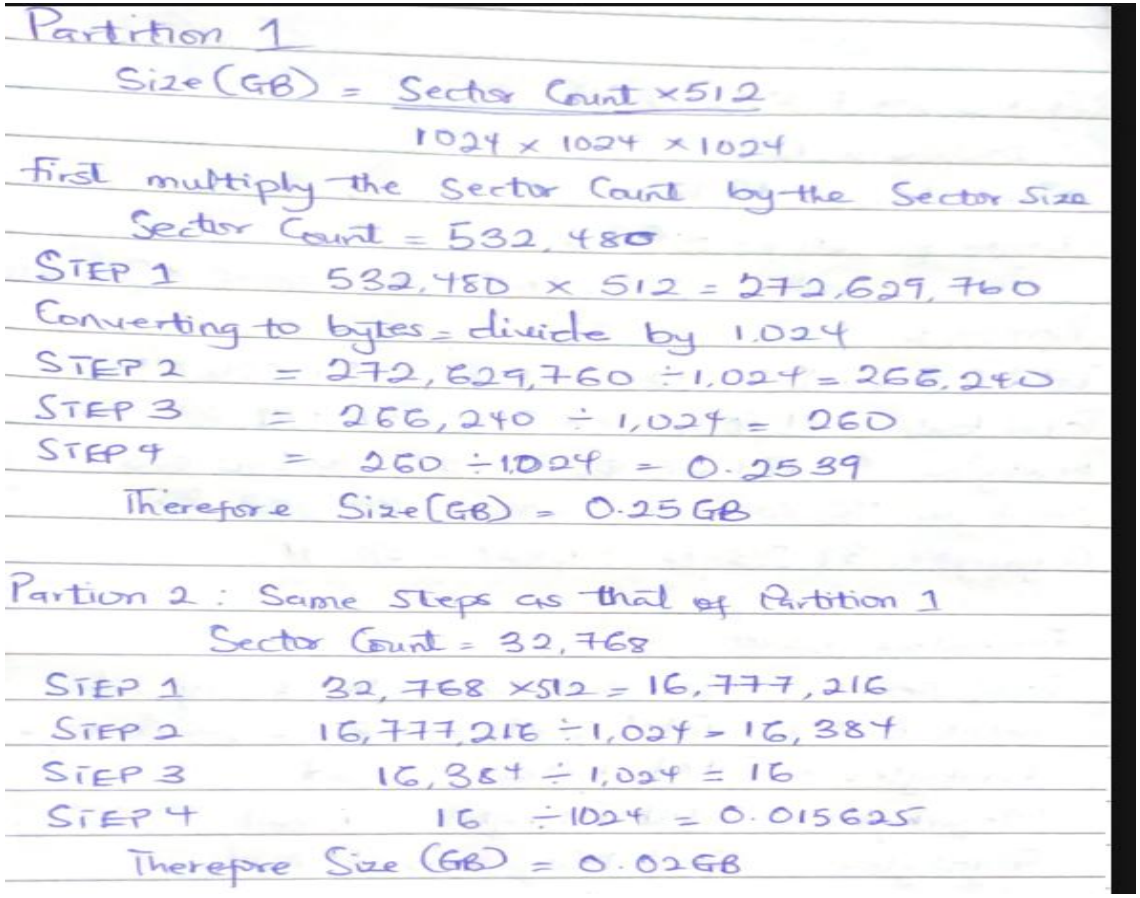
- For each partition, record the start sector, number of sectors, sector size, and file system where available.



- Calculate the size of each partition in GB using the formula above.
- Your calculation must be shown step-by-step. Screenshots alone are not enough.

> Example calculation:
> Partition size in GB = sector_count x 512 / 1024 / 1024 / 1024

If sector_count = 83,881,984:
83,881,984 x 512 / 1024 / 1024 / 1024 = 39.99 GB



Partition 1

Size (GB) = $\frac{\text{Sector Count} \times 512}{1024 \times 1024 \times 1024}$

First multiply the Sector Count by the Sector Size

Sector Count = 532,480

STEP 1 532,480 × 512 = 272,629,760

Converting to bytes = divide by 1,024

STEP 2 = 272,629,760 ÷ 1,024 = 266,240

STEP 3 = 266,240 ÷ 1,024 = 260

STEP 4 = 260 ÷ 1024 = 0.2539

Therefore Size (GB) = 0.25 GB

Partion 2 : Same steps as that of Partition 1

Sector Count = 32,768

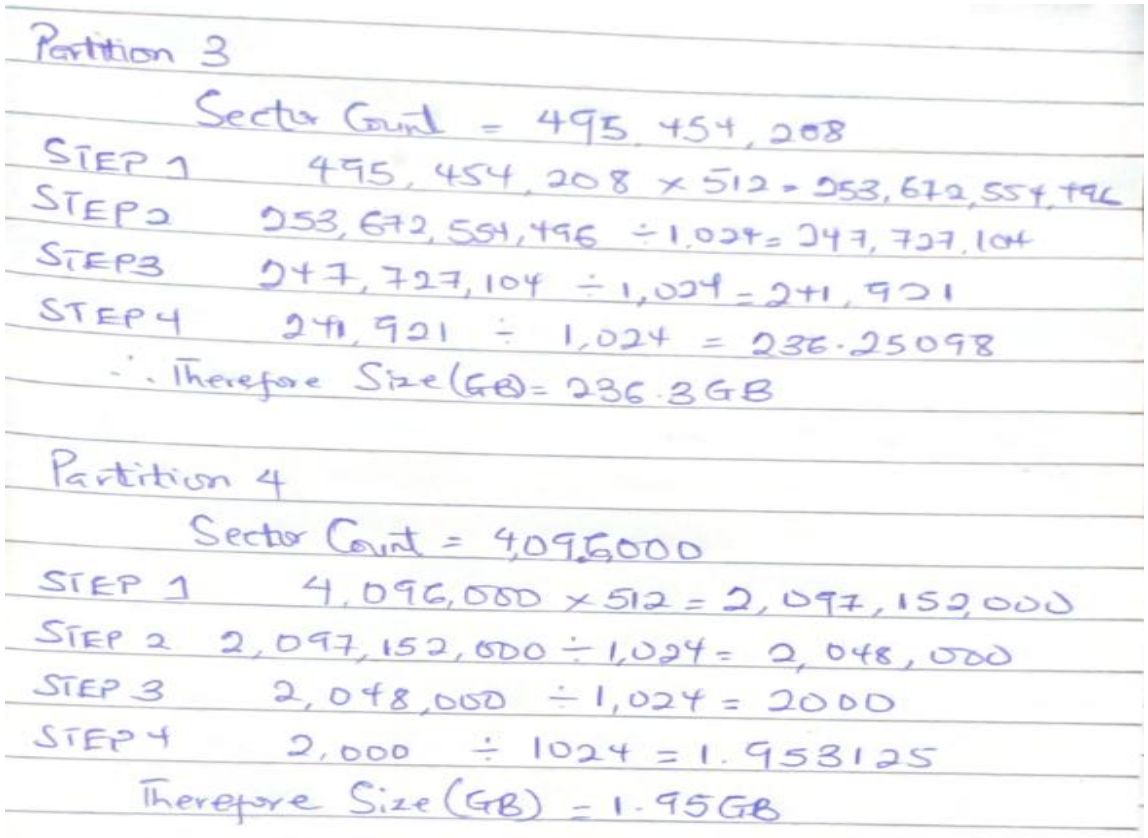
STEP 1 32,768 × 512 = 16,777,216

STEP 2 16,777,216 ÷ 1,024 = 16,384

STEP 3 16,384 ÷ 1,024 = 16

STEP 4 16 ÷ 1024 = 0.015625

Therefore Size (GB) = 0.02 GB

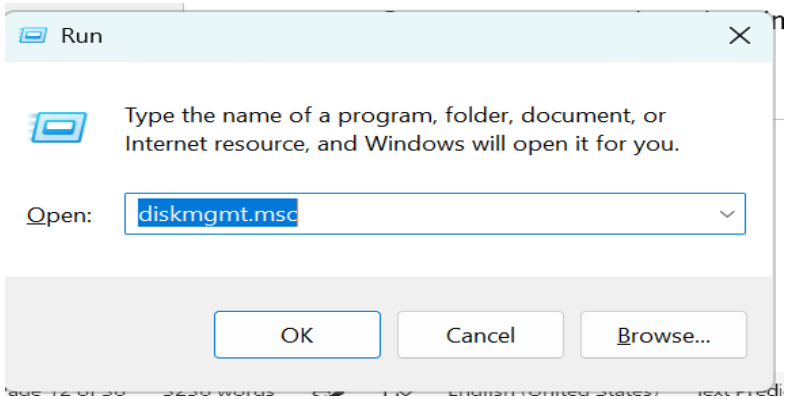


Part D - Safe Partition Table Inspection

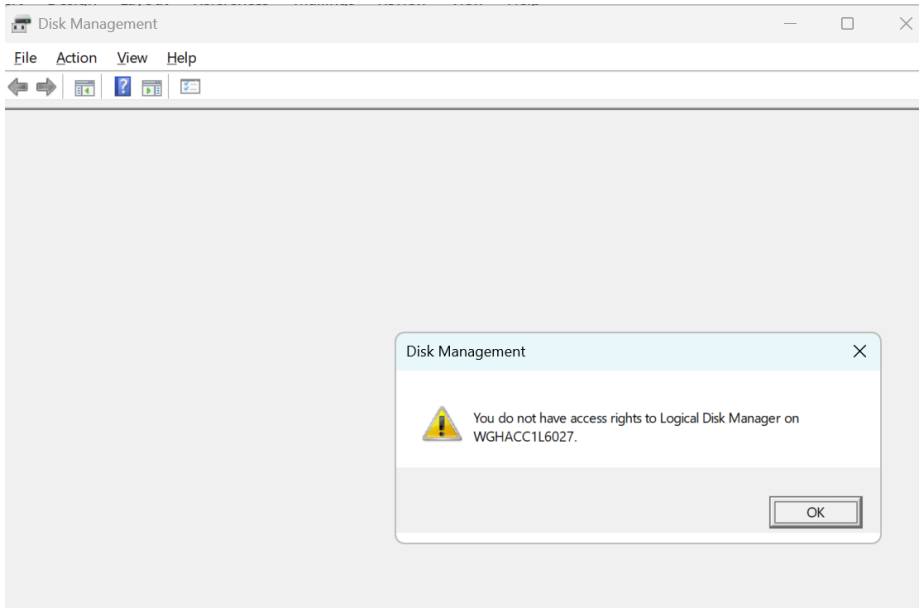
D1. Windows method - Disk Management, PowerShell and DiskPart

- Open Disk Management by pressing Windows + R, typing diskmgmt.msc, and pressing Enter.

Capture a screenshot showing your disk layout and visible partitions

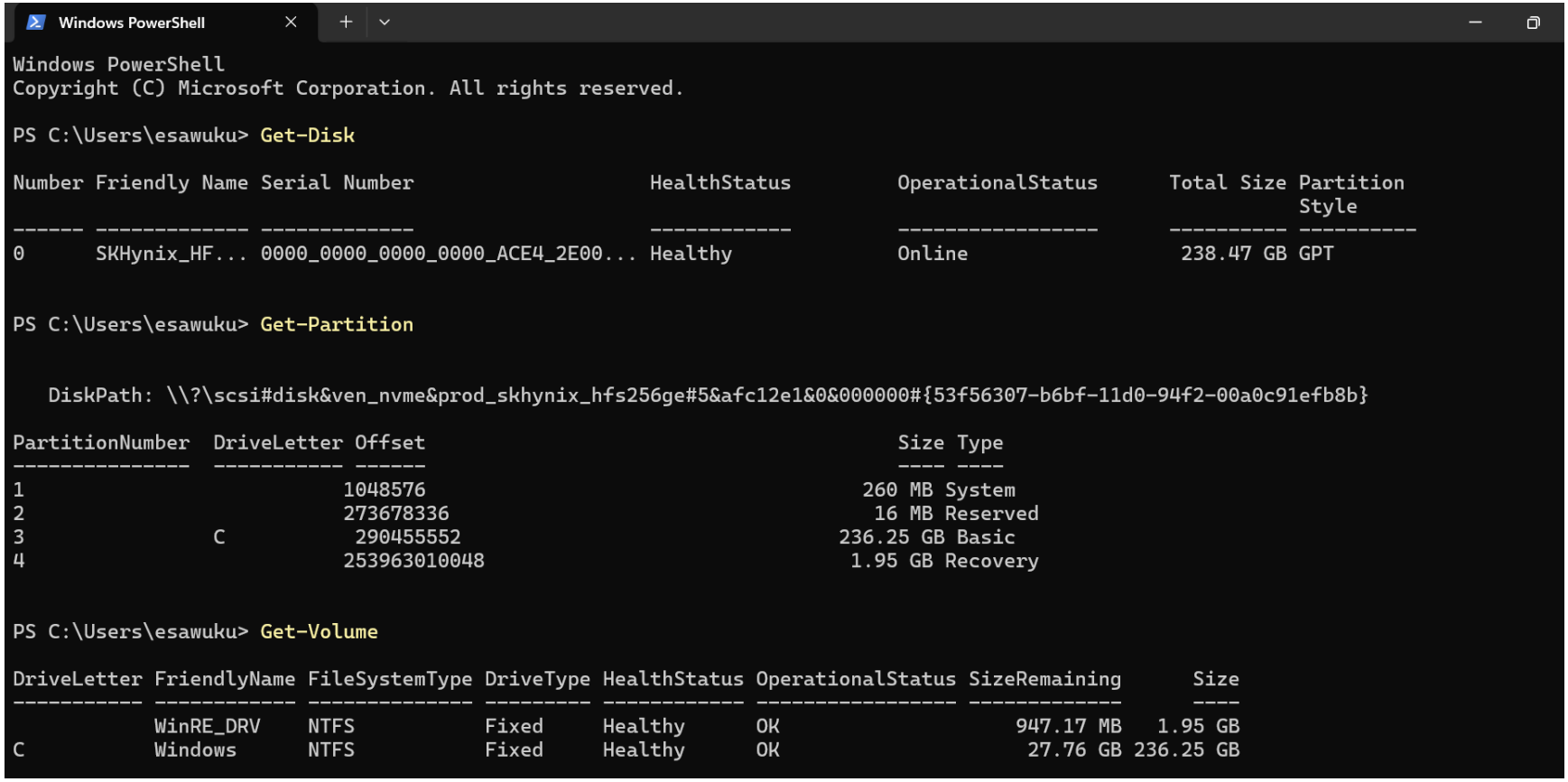


A per the policy on the machine I am using, I do not have access to view



- Open PowerShell as normal user or administrator if required and run the safe read-only commands below.

Powershell : (Get- Disk, Get-Partition, Get-Volume)

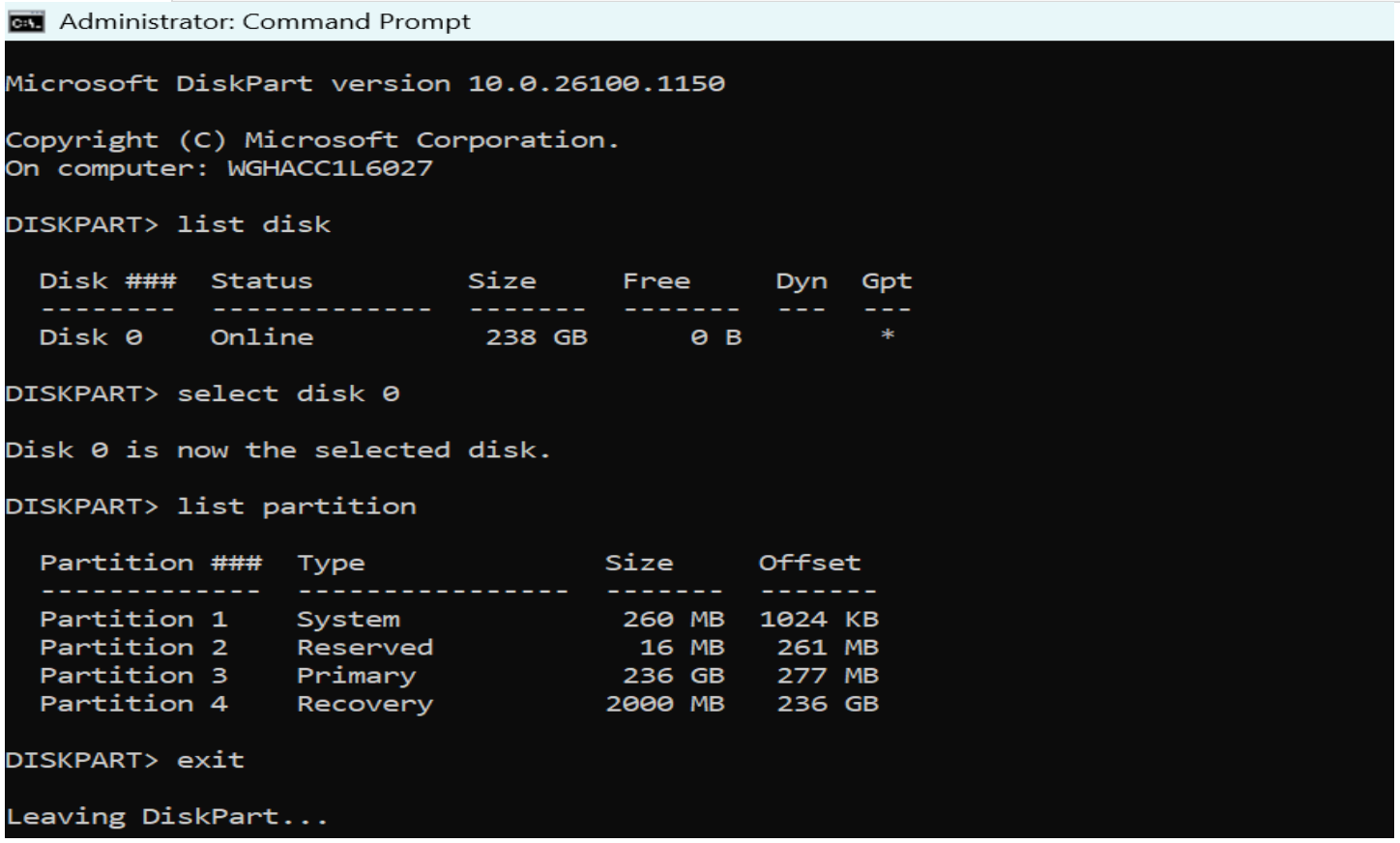


- Open Command Prompt as administrator and run DiskPart only for listing disks and partitions. Do not run create, delete, clean, format, active, or convert commands.

```
PowerShell:
Get-Disk
Get-Partition
Get-Volume
```

DiskPart safe listing only:

```
diskpart
list disk
select disk 0
list partition
exit
```



D2. Linux/Kali method - read-only inspection

- Open Terminal in your Linux/Kali VM or host.
- Run the commands below and capture screenshots showing disk, partition, size, and file system information.
- Do not run mkfs, fdisk write commands, parted write commands, dd write commands, or mount read-write commands for this lab.

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
sudo fdisk -l
sudo parted -l
blkid
df -h
```

The below commands ***(lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT, sudo fdisk -1 ,sudo parted -1, blkd and df -h),*** the first command displays all the disks and partitions, the second command displays the detailed disk and partition information, whiles the last command displays the partition information using parted. The fourth command displays the file system and the last command displays the mounted file systems and the available space left.

```
┌──(icdfa㉿icdfa)-[~]
└─$ lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINT
NAME    SIZE TYPE FSTYPE MOUNTPOINT
sda      50G disk
├─sda1 41.7G part ext4   /
├─sda2    1K part
├─sda3    7G part ext4
└─sda5  1.3G part swap   [SWAP]

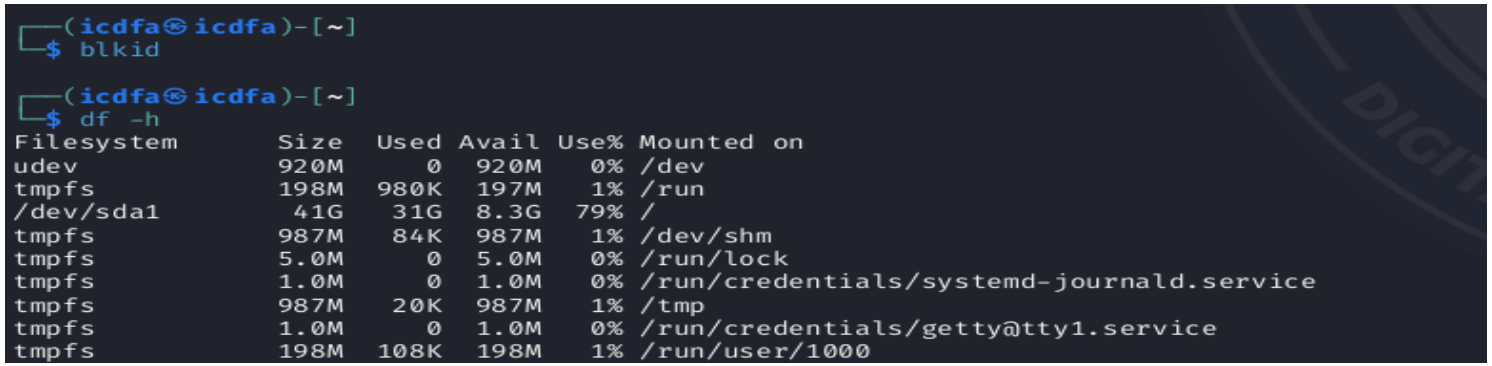
┌──(icdfa㉿icdfa)-[~]
└─$ sudo fdisk -l
[sudo] password for icdfa:
Disk /dev/sda: 50 GiB, 53687091200 bytes, 104857600 sectors
Disk model: VBOX HARDDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x1bece6b5

Device     Boot    Start       End   Sectors  Size Id Type
/dev/sda1  *        2048  87390207 87388160 41.7G 83 Linux
/dev/sda2       87390208  90175487  2785280  1.3G  f W95 Ext'd (LBA)
/dev/sda3       90175488 104857599 14682112    7G 83 Linux
/dev/sda5       87392256  90175487  2783232  1.3G 82 Linux swap / Solaris

Partition table entries are not in disk order.

┌──(icdfa㉿icdfa)-[~]
└─$ sudo parted -l
Model: ATA VBOX HARDDISK (scsi)
Disk /dev/sda: 53.7GB
Sector size (logical/physical): 512B/512B
Partition Table: msdos
Disk Flags:

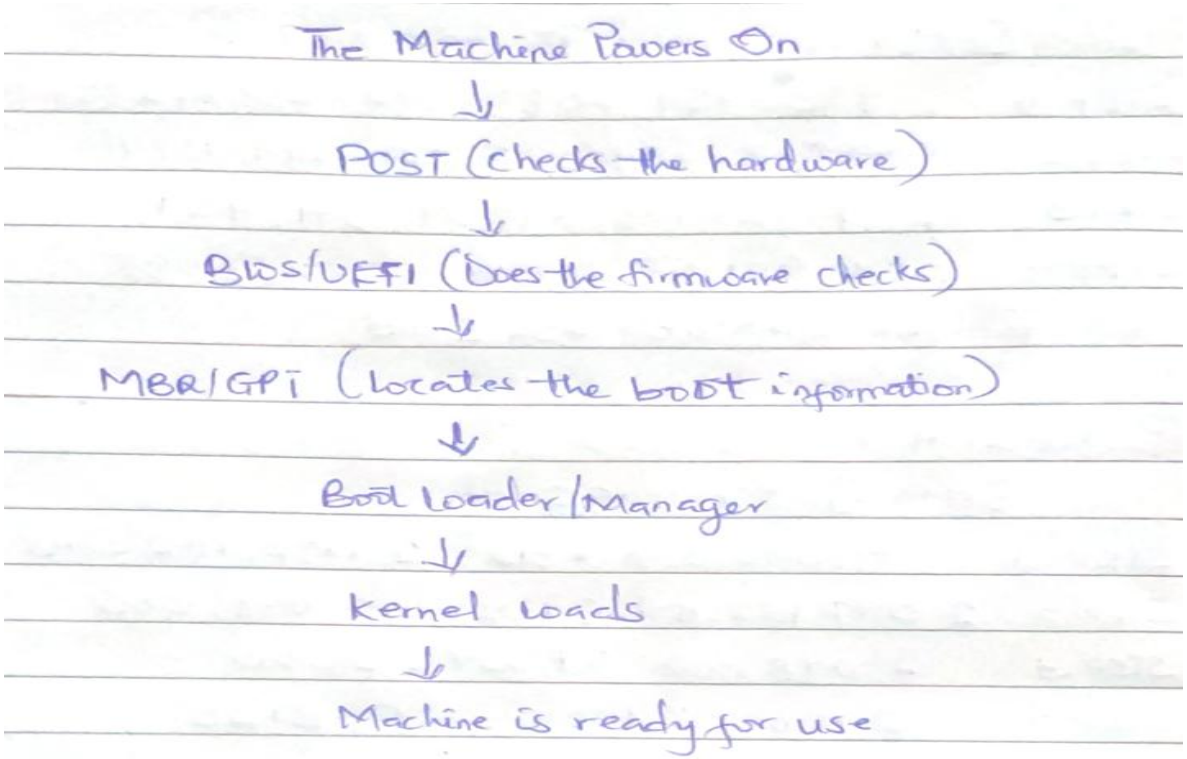
Number  Start   End     Size    Type      File system     Flags
 1      1049kB  44.7GB  44.7GB  primary   ext4            boot
 2      44.7GB  46.2GB  1426MB  extended                  lba
 5      44.7GB  46.2GB  1425MB  logical   linux-swap(v1)  swap
 3      46.2GB  53.7GB  7517MB  primary   ext4
```



Part E - MBR, GPT and Boot Process Explanation

E1. Explain the boot process

- Create a simple flow diagram or numbered list explaining how a PC boots.



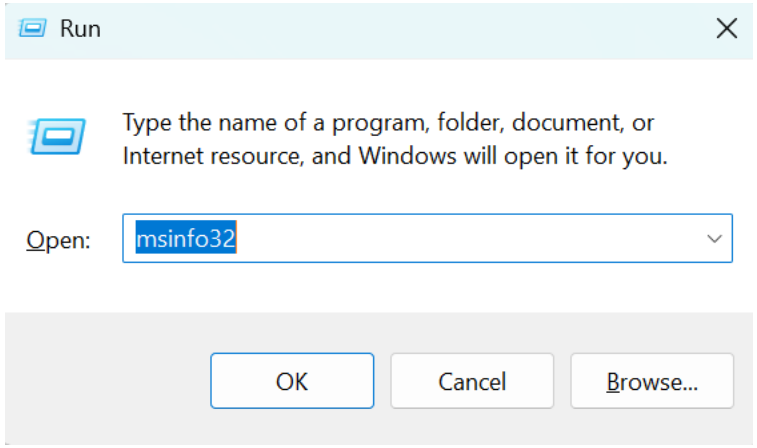
- Your explanation must include power-on, POST, BIOS/UEFI, CMOS/firmware settings, MBR/GPT or EFI boot records, boot loader/manager, kernel loading, and operating system control.

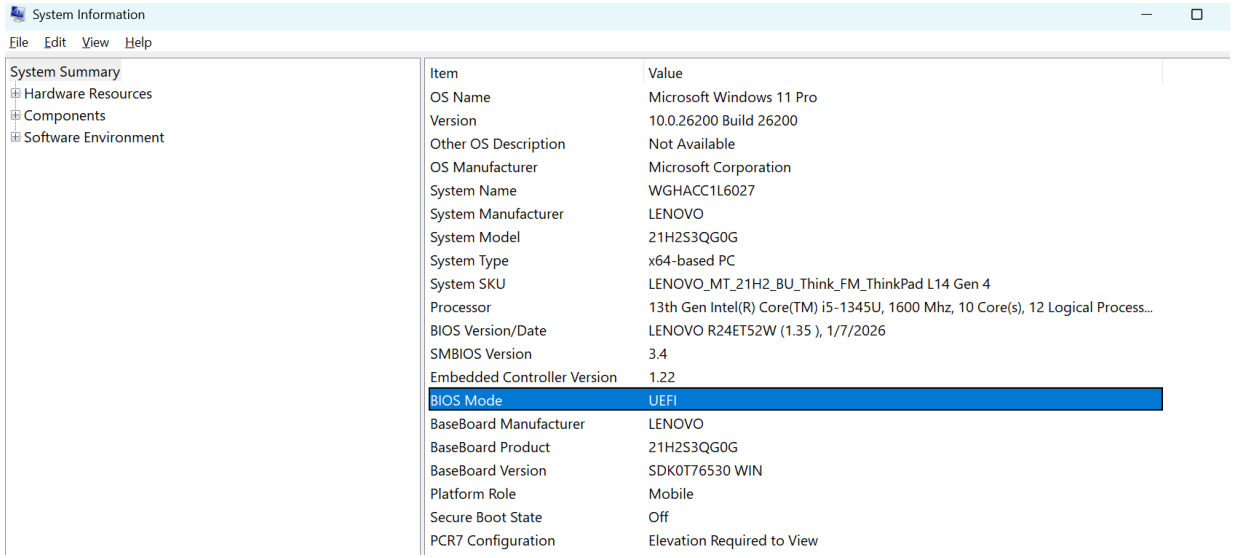
Stage	What Happens	Forensic Relevance
Power-on and POST	Once I power on the machine, the machine performs a Power-On-Self-Test (POST), what it does is checking if all hardware components on the machine such as the processor, RAM and other hardware components are working perfectly	If there is any tampering with the hardware, the computer might not start up correctly and may tamper forensic analysis.

BIOS/UEFI firmware	The BIOS/UEFI happens to be the computer's built in startup program, checking the settings of the firmware and the boot order sequence and then decides how and where to start the operating system	If there are any changes made to the boot settings, there would obviously be some changes that have been made to the system's startup configuration.
MBR/GPT or EFI boot	What happens here is that the computer uses the boot method to locate the information needed to start the OS. The Master Boot Record (MBR) is mostly used by the legacy BIOS while UEFI uses boot entries and files stored in an EFI system partition, which is also usually on the GPT disk.	The boot records and the EFI files can provide some evidence about the computer's boot configuration and the installed OS.
Boot loader/manager	The Boot loader in the machine helps in loading the files that are needed to start the OS whiles the boot manager can also allow the user to choose which OS is needed	The boot loader/manager has some multiple configurations that are of relevance during forensic investigations.
Kernel and OS control	This is the main part of the OS, it controls the computer, starting the services and preparing the computer for the User to use it.	As the OS begins to start, there are some user activities and system logs that are captured for forensic analysis.

State whether your own system uses BIOS/Legacy or UEFI based on system information evidence.

My System uses a UEFI as its boot mode. This initializes my computer’s hardware and then locates the boot loader that helps start my computer’s OS.

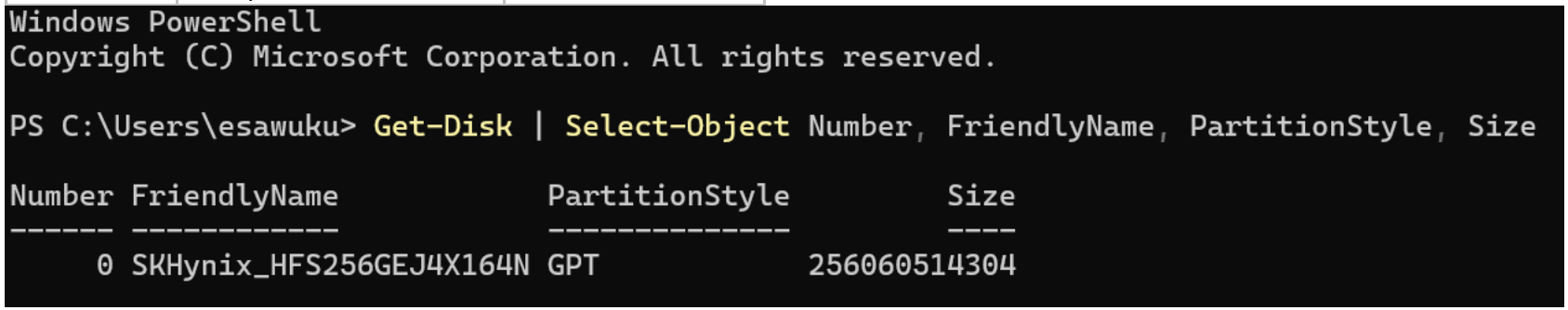




E2. Compare MBR and GPT

Feature	MBR	GPT

Age / design	Older partitioning system from the 1980s.	Modern partitioning system is used with UEFI systems.
Size limitation	Commonly limited to about 2 TB.	Supports very large disks; slide notes up to about 9.4 ZB.
Partition information	Partition table is stored in the first sector with boot code.	Uses primary and backup GPT headers and partition entries.
Forensic issue	Critical information in sector 0; damage can affect partition interpretation.	Backup structures may help recovery and validation.



My computer uses GPT partition Style using PowerShell to check my partition style on my computer's disk. The computer also uses a UEFI boot mode. With this evidence I can clearly say from the table that my computer uses GPT, which is a modern GPT partition rather than the older, which is MBR method.

Part F - File System Identification and Interpretation

- Identify the file system used by at least two volumes or partitions on your system.

```
PS C:\Users\esawuku> Get-Volume

DriveLetter FriendlyName FileSystemType DriveType HealthStatus OperationalStatus SizeRemaining      Size
----------- ------------ -------------- --------- ------------ ----------------- -------------      ----
            WinRE_DRV    NTFS           Fixed     Healthy      OK                    947.17 MB   1.95 GB
C           Windows      NTFS           Fixed     Healthy      OK                     27.87 GB 236.25 GB
```

- For Windows systems, identify whether volumes use NTFS, FAT/FAT32, exFAT, or another file system.
  My windows system uses New File Technology System ***(NTFS)***, both volumes on my computer use NTFS.

- For Linux systems, identify whether partitions use ext4, XFS, btrfs, swap, or another file system.
  The Linux system uses ext4, which is a fourth extended filesystem which is commonly used by Linux to organize and store files. It also has Swap which is a special disk space Linux uses to help support the memory management when the physical RAM is not enough.

```
┌──(icdfa㉿icdfa)-[~]
└─$ lsblk -f
NAME   FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1 ext4   1.0         a5089483-4c45-4934-865d-711741236ab9    8.3G    75% /
├─sda2
├─sda3 ext4   1.0         5653e517-df4a-42ef-bbb0-9fd4344df273
└─sda5 swap   1           ddf8b4fe-2814-4bd7-91e3-ff149998eba2                [SWAP]

┌──(icdfa㉿icdfa)-[~]
└─$ blkid

┌──(icdfa㉿icdfa)-[~]
└─$ df -Th
Filesystem     Type      Size  Used Avail Use% Mounted on
udev           devtmpfs  920M     0  920M   0% /dev
tmpfs          tmpfs     198M  976K  197M   1% /run
/dev/sda1      ext4       41G   31G  8.3G  79% /
tmpfs          tmpfs     987M  4.0K  987M   1% /dev/shm
tmpfs          tmpfs     5.0M     0  5.0M   0% /run/lock
tmpfs          tmpfs     1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs          tmpfs     987M  8.0K  987M   1% /tmp
tmpfs          tmpfs     1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs          tmpfs     198M  108K  198M   1% /run/user/1000
```

- In your report, explain why a file system is important in digital forensics.

File System is very important in digital forensics because it really helps during forensic Analysis, how files are been stored and saved and can track down files that have been deleted and be able to conduct investigations during an incident and recover them.

Part G - Final Practical Assignment

Screenshot G1

Get-Disk | select-object Number, FriendlyName, PartitionStyle, LogicalSectorSize, Size

```
PS C:\WINDOWS\system32> Get-Disk | select-object Number, FriendlyName, PartitionStyle, LogicalSectorSize, Size


Number            : 0
FriendlyName      : KBG6AZNV256G LA KIOXIA
PartitionStyle    : GPT
LogicalSectorSize : 512
Size              : 256060514304



PS C:\WINDOWS\system32> Get-Partition -DiskNumber 0 | Format-Table DiskNumber, PartitionNumber, DriverLetter, Offset, Size, Type -AutoSize

DiskNumber PartitionNumber DriverLetter       Offset         Size Type
---------- --------------- ------------       ------         ---- ----
         0               1                    1048576    209715200 System
         0               2                  210763776     16777216 Reserved
         0               3                  227540992 255013683200 Basic
         0               4             255241224192    817889280 Recovery
```

Screenshot G2

Get-Partition | Select-Object DiskNumber, PartitionNumber, @{Name="Start Sector";Expression={$_.Offset/512}}, @{Name="Sector Count";Expression={$_.Size/512}}, @{Name="End Sector";Expression={($_.Offset/512)+($_.Size/512)-1}} | Format-Table -AutoSize

```
PS C:\WINDOWS\system32> Get-Partition | Select-Object DiskNumber, PartitionNumber, @{Name="Start Sector";Expression={$_.Offset/512}}, @{Name="Sector Count";Expression={$_.Size/512}}, @{Name="End Sector";Expres
sion={($_.Offset/512)+($_.Size/512)-1}} | Format-Table -AutoSize

DiskNumber PartitionNumber Start Sector Sector Count End Sector
---------- --------------- ------------ ------------ ----------
         0               1         2048       409600     411647
         0               2       411648        32768     444415
         0               3       444416    498073600  498518015
         0               4    498518016      1597440  500115455
```

Screenshot G3

Partition Calculation in GB: $\frac{\text{Sector Count} * 512}{1024 * 1024 * 1024}$

Calculation done for all partitions

Partition 1

Size GB = Sector Count × 512
1,024 × 1,024 × 1,024

We first multiply the Sector Count by Sector Size

Sector Count = 409600

Step 1 = 409600 × 512 = 209,715,200

Step 2 = Divider by 1024

209,715,200 ÷ 1024 = 204,800KB

Step 3 = 204,800 ÷ 1024 = 200MB

Step 4 = 200 ÷ 1024 = 0.1953GB

Therefore Size GB = ~~0.0GB~~ 0.1953GB

Partition 2

Size GB = Sector Count × 512
1,024 × 1,024 × 1,024

Sector Count = 32768

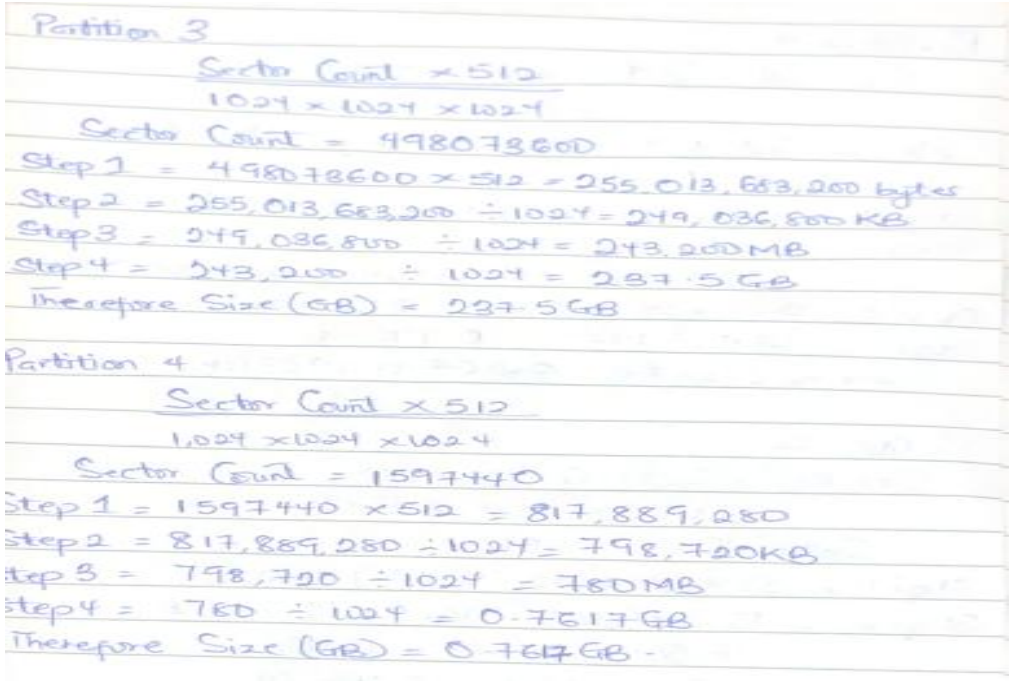
Step 1 = 32768 × 512 = 16,777,216

Step 2 = 16,777,216 ÷ 1024 = 16,384KB

= 16,384 ÷ 1024 = 16MB

= 16 ÷ 1024 = 0.0156 GB

Therefore Size (GB) = ~~0.00GB~~ 0.0156GB



Screenshot G4

```
PS C:\WINDOWS\system32> Get-Partition -DiskNumber 0 | ForEach-Object {
>>     $p = $_
>>     $v = if ($p.DriveLetter) { Get-Volume -DriveLetter $p.DriveLetter -ErrorAction SilentlyContinue }
>>     [PSCustomObject]@{
>>         PartitionNumber = $p.PartitionNumber
>>         DriveLetter = $p.DriveLetter
>>         Type = $p.Type
>>         PartitionSizeGB = [math]::Round($p.Size/1GB,4)
>>         PartitionSizeMB = [math]::Round($p.Size/1MB,2)
>>         VolumeSizeGB = if ($v) {[math]::Round($v.Size/1GB,4)} else {"N/A"}
>>         VolumeSizeMB = if ($v) {[math]::Round($v.Size/1MB,2)} else {"N/A"}
>>         RemainingGB = if ($v) {[math]::Round($v.SizeRemaining/1GB,4)} else {"N/A"}
>>     }
>> } | Format-Table -AutoSize

PartitionNumber DriveLetter Type     PartitionSizeGB PartitionSizeMB VolumeSizeGB VolumeSizeMB RemainingGB
--------------- ----------- ----     --------------- --------------- ------------ ------------ -----------
              1             System           0.1953             200 N/A          N/A          N/A
              2             Reserved         0.0156              16 N/A          N/A          N/A
              3            C Basic             237.5          243200 237.5        243200       165.3054
              4             Recovery         0.7617             780 N/A          N/A          N/A


PS C:\WINDOWS\system32> Get-Partition -DiskNumber 0 | Where-Object {$_.PartitionNumber -in 1,2} | ForEach-Object {
>>     $p = $_
>>     $v = if ($p.DriveLetter) { Get-Volume -DriveLetter $p.DriveLetter -ErrorAction SilentlyContinue }
>>     [PSCustomObject]@{
>>         PartitionNumber = $p.PartitionNumber
>>         Type = $p.Type
>>         PartitionSizeGB = [math]::Round($p.Size/1GB,4)
>>         PartitionSizeMB = [math]::Round($p.Size/1MB,2)
>>         VolumeSizeGB = if ($v) {[math]::Round($v.Size/1GB,4)} else {"N/A"}
>>         VolumeSizeMB = if ($v) {[math]::Round($v.Size/1MB,2)} else {"N/A"}
>>         RemainingSpaceGB = if ($v) {[math]::Round($v.SizeRemaining/1GB,4)} else {"N/A"}
>>     }
>> } | Format-Table -AutoSize

PartitionNumber Type     PartitionSizeGB PartitionSizeMB VolumeSizeGB VolumeSizeMB RemainingSpaceGB
--------------- ----     --------------- --------------- ------------ ------------ ----------------
              1 System           0.1953             200 N/A          N/A          N/A
              2 Reserved         0.0156              16 N/A          N/A          N/A
```

Task	Required Evidence	Student Notes / Answer
Identify partition 1	***Screenshot G1*** ***Get-Disk | select-object Number, FriendlyName, PartitionStyle, LogicalSectorSize, Size***	This command was run on PowerShell to get the partitions, the sector sizes and type of the partition.

Record partition 1 sector count	***Screenshot G2,*** ***Get-Partition | Select-Object DiskNumber, PartitionNumber, @{Name="Start Sector";Expression={$_.Offset/512}},@{Name="Sector Count";Expression={$_.Size/512}}, @{Name="End Sector";Expression={($_.Offset/512)+($_.Size/512)-1}} | Format-Table -AutoSize***	This command was run on PowerShell to get the start sector and the end sector, to get the sector count to calculate for the size in GB
Calculate partition 1 size	***Screenshot G3, 0.1953GB***	Sector Size was divided by ***(1,024 * 1,024 *1,024)*** to get the size in GB
Identify partition 2	***Screenshot G1*** ***Get-Disk | select-object Number, FriendlyName, PartitionStyle, LogicalSectorSize, Size***	This command was run on PowerShell to get the partitions, the sector sizes and type of the partition.
Record partition 2 sector count	***Screenshot G2,*** ***Get-Partition | Select-Object DiskNumber, PartitionNumber, @{Name="Start Sector";Expression={$_.Offset/512}},@{Name="Sector Count";Expression={$_.Size/512}}, @{Name="End Sector";Expression={($_.Offset/512)+($_.Size/512)-1}} | Format-Table -AutoSize***	This command was run on PowerShell to get the start sector and the end sector, to get the sector count to calculate for the size in GB
Calculate partition 2 size	***Screenshot G3, 0.0156GB***	Sector Size was divided by ***(1,024 * 1,024 *1,024)*** to get the size in GB
Compare with OS-reported size	**Screenshot G4**	I realized that the size of GB was the same when I run the command in PowerShell, however the volume for partition 1 and 2 was not available (N/A) because they are EFI and Microsoft reserved partitions without a normal accessible volume

Explain difference if any	Short explanation of rounding, reserved space, file system overhead, or GiB/GB differences	There are differences because the small differences in partition sizes may be because of rounding or the way the storage is measured uses ***1,073,741,824 bytes*** while GB uses ***1,000,000,000 bytes***. We can see that one partition that is Partition 2 is reserved because some machines require a reserved space for Windows system operations and the file system also uses a small amount of space when it comes to managing stored data

Conclusion

This practical exercise carried out has helped me to successfully understand and identify Partitions and the sizes of those sizes uses PowerShell. Getting the Start Sectors, Calculating the sector counts and comparing with the windows-reported values that shows.

References

1. 1. Microsoft. (n.d.). Get-Partition (Storage). Microsoft Learn. https://learn.microsoft.com/en-us/powershell/module/storage/get-partition
2. Microsoft. (n.d.). DiskPart. Microsoft Learn. https://learn.microsoft.com/en-us/windows-server/administration/windows-commands/diskpart
3. Microsoft. (n.d.). Get-Volume (Storage). Microsoft Learn. https://learn.microsoft.com/en-us/powershell/module/storage/get-volume
4. Slides from ICDFA.